

Many Steps of Evaluation: $\rightarrow^*$ for the reflexive transitive closure of $\rightarrow$:
 $P, S, M \rightarrow^* P', S', M' \Leftrightarrow (P, S, M) = (P', S', M') \vee \exists P'', S'', M''. P, S, M \rightarrow P'', S'', M'' \wedge P'', S'', M'' \rightarrow^* P', S', M'.$

SC Traces:

Initial Memory: $M_0 \triangleq \lambda x. 0$. **Initial Store:** $s_0 \triangleq \lambda a. 0$. **Initial Store Map:** $S_0 \triangleq \lambda \tau. s_0$. **Terminated Program:** $P_{skip} \triangleq \lambda \tau. skip$.

Initial SC-configuration: (P, S_0, M_0) .

SC-trace: an evaluation path that starts from the initial SC-configuration of P and terminates with P_{skip} . i.e. there exists S and memory M such that: $P, S_0, M_0 \rightarrow^* P_{skip}, S, M$.

A pair (S, M) denotes an **SC-outcome** iff $P, S_0, M_0 \rightarrow^* P_{skip}, S, M$.

SC is not deterministic nor confluent.

Deterministic: